Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	01-10-26
Team ID	23KD1A4230
Project Name	Metro Ticket Generating System in ServiceNow
Maximum Marks	4 Marks

Technical Architecture:

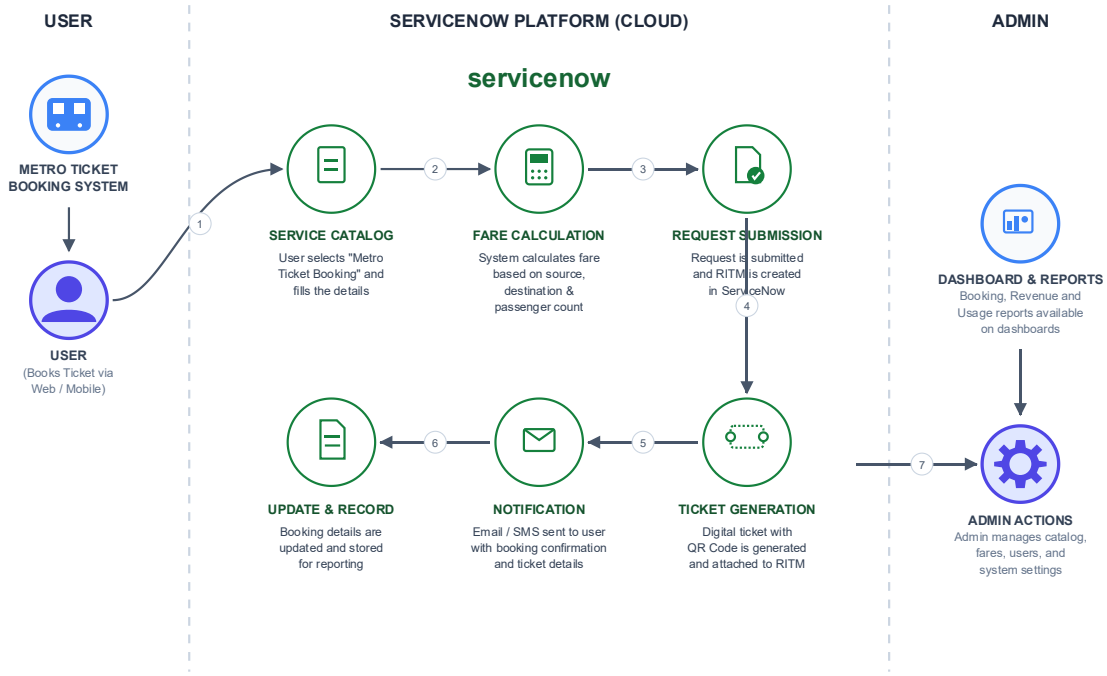


Table-1: Components & Technologies

S.No	Component	Description	Technology
1.	User Interface	Interface for passengers to book metro tickets	ServiceNow Service Catalog, Portal UI

S.No	Component	Description	Technology
2.	Application Logic-1	Fare calculation based on source and destination stations	Catalog Client Scripts (JavaScript)
3.	Application Logic-2	Server-side fare retrieval and validation	Script Includes, GlideAjax
4.	Application Logic-3	Automated ticket booking workflow and notifications	Flow Designer, Business Rules
5.	Database	Stores metro stations, fares, and booking information	ServiceNow Tables (u_metro_station, u_metro_fare, sc_req_item)
6.	Cloud Database	Cloud-based data storage managed by ServiceNow	ServiceNow Cloud Database
7.	File Storage	Stores QR codes, ticket attachments, and records	ServiceNow Attachments
8.	External API-1	QR Code generation for digital tickets	QR Code Generator API / Scripted QR Generation
9.	External API-2	Email notification service for ticket confirmation	ServiceNow Notification Engine
10.	Machine Learning Model	Ticket booking trends, revenue, and usage reports	ServiceNow Reports & Dashboards
11.	Infrastructure (Server / Cloud)	Application hosting and execution environment	ServiceNow SaaS Cloud Platform

Table-2: Application Characteristics

S.No	Characteristics	Technology	Description
1.	Open-Source Frameworks	JavaScript, GlideAjax	Client-side scripting and customization used for fare calculation and form validation
2.	Security Implementations	ServiceNow ACLs, User Roles, HTTPS, IAM Controls	Role-based access control for administrators and users, authenticated access through ServiceNow platform, secure data storage and transmission
3.	Scalable Architecture	ServiceNow SaaS Architecture, Multi-Tenant Cloud Platform	ServiceNow's cloud-based multi-tenant architecture supports growing users and ticket bookings without infrastructure changes
4.	Availability	ServiceNow Cloud	High availability through ServiceNow cloud infrastructure with continuous monitoring and managed services
5.	Performance	GlideAjax, Script Includes, ServiceNow Database	Fast ticket booking process through server-side processing with optimal database queries, and cloud-hosted services

References

- https://www.servicenow.com/docs/r/servicenow-platform/service-catalog/service-catalog.html
- https://www.servicenow.com/docs/r/application-development/building-applications/flow-designer.html
- https://www.servicenow.com/docs/r/api-reference/c_GlideAjaxAPI.html
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